

Test 19: Dirac cone $v_F \approx 0.125$

Verdict: PASS | Generated: 2026-09-15 01:24:34 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

What this test verifies

EXACT-geometry: zero modes at $(\pi/2, \pi/2)$ require $L/4 \gg \lambda$; gap scaling fit $R \approx 1$. FOUR-pass (user spec 2026-09-04): primary fit ≈ 11 -pt hardcore audit $\approx 10^{-10}$ -series $\approx 10^{-10}$ -twist-spectroscopy $\approx 10^{-10}$ -eigenvalue index

METHOD: EXACT-geometry: zero modes at $(\pi/2, \pi/2)$ require $L/4 \gg \lambda$; gap scaling fit $R \approx 1$. FOUR-pass (user spec 2026-09-04): primary fit ≈ 11 -pt hardcore audit $\approx 10^{-10}$ -series $\approx 10^{-10}$ -twist-spectroscopy $\approx 10^{-10}$ -eigenvalue index

Test 19 AUDIT: v_F _twist = 1.9990 ($R \approx 1.00000$, $L=48$ y-twist scan), tower 4 @ 10^{-10} with 2+2 polarization, 2 π -holonomy 2.5×10^{-17} , π -cusp 6.279, rebuild bit-exact, $1/L \approx 0.021$ intercept 1.00000

Result [AUDIT pass 4]: 8 sub-checks, 0 failed $\approx$ PASS

- Plots
- 'plots/plot_01.png' $\approx$ 1600x1000, ABPlotV23 engine (supersampled)
 - 'plots/plot_02.png' $\approx$ 1600x1000, ABPlotV23 engine (supersampled)
 - 'plots/plot_03.png' $\approx$ 1600x1000, ABPlotV23 engine (supersampled)
 - 'plots/animation.gif' $\approx$ animated reveal of every plot of this test

Reproduce with: `julia ab_cloud_v23.jl --test 19 --no-two-pass`

Every computation is logged in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.

Independent verification guide

[01:23:43.346] [+5814.374s] build_ab_cloud_hamiltonian: 48x48 (N=2304), $\lambda \approx 0.5000$, $t=1.00$, $W=0.00$, torus, 0 vortices, model=:monumental

[01:24:31.332] [+5862.360s] build_ab_cloud_hamiltonian: 48x48 (N=2304), $\lambda \approx 0.5000$, $t=1.00$, $W=0.00$, torus, 0 vortices, model=:monumental

FULL COMPUTATION LOG

??

TEST 19 (pass 4, AUDIT): Dirac cone $\approx$ twist spectroscopy + eigenvector index

CONSOLE CAPTURE

Direct $E(k)$ at the Dirac point (pure-holonomy y-twist scan), All index of

the tower, 2 π -holonomy, determinism, $1/L \approx 0.021$ collapse, cross-pass v_F audit

??

twist lattice 48x48 ($\lambda \approx 1/2$, torus, $W=0$, clean), π -grid $\pi/5 \approx 0.628$ + $\lambda \approx 0$ anchor

$\approx$ RUNTIME: 7 dense solves at 48 π (2304 sites) + 2 builds $\approx 1 \pi$ min

$\lambda \approx 0$: tower($|E| \approx 10^{-10}$) = 4, $E_{\min} = 0.261052$ ($E_{\min} \cdot L = 12.5305$; pass-2 measured 12.5305)

$\lambda \approx 0.6283$: $E_{\min} = 0.026179$ ($E_{\min} \cdot L = 1.2566$), $n_{\text{zero}} = 0$, degeneracy = 2, v_F _local = 1.9999

$\lambda \approx 1.2566$: $E_{\min} = 0.052354$ ($E_{\min} \cdot L = 2.5130$), $n_{\text{zero}} = 0$, degeneracy = 2, v_F _local = 1.9998

$\lambda \approx 1.8850$: $E_{\min} = 0.078520$ ($E_{\min} \cdot L = 3.7689$), $n_{\text{zero}} = 0$, degeneracy = 2, v_F _local = 1.9995

$\lambda \approx 2.5133$: $E_{\min} = 0.104672$ ($E_{\min} \cdot L = 5.0243$), $n_{\text{zero}} = 0$, degeneracy = 2, v_F _local = 1.9991

$\lambda \approx 3.1416$: $E_{\min} = 0.130806$ ($E_{\min} \cdot L = 6.2787$), $n_{\text{zero}} = 0$, degeneracy = 4, v_F _local = 1.9986

D1 no-intercept fit: $E = 0.041645 \cdot v_F$, $R^2 = 1.000000$? v_F twist = 1.9990 ?

D1b v_F twist vs pass-2 fit: 1.9990 vs 1.9447 ($|v_F| < 5\%$) ? OK

D2 tower polarization: $\text{Tr}(P_A P_T) = 2.0000000000$, $\text{Tr}(P_B P_T) = 2.0000000000$ (expect 2, 2 ? chirality-balanced Dirac pair, AIII index 0) ? OK

per-vector purity (LAPACK basis, diagnostic only): 0.500, 0.499, 0.499, 0.500

D3a 2D-holonomy: $\langle H(2) \rangle \langle H(0) \rangle \langle H \rangle = 2.50e-17$, $\max|E| = 4.35e-14$ (Byers-Yang at the twist level) ? OK

D3b ?= ? cusp: $n_{\text{zero}} = 0$ (tower annihilated), $E_{\text{min}} L = 6.2787$? [5.0, 7.6] (target 2 = 6.283; degeneracy 4 ? both cones meet at the cusp) ? OK

D4 determinism: fresh $W=0$ rebuild ? $\langle H \rangle \langle F \rangle \langle H \rangle = 0.0e+00$ (bit-exact expected ? clean fast path draws no RNG) ? OK

D5 1/L ? collapse (11 pts, pass-2 series): $E_{\text{min}} L = 12.5662 + (-82.06)/L$?, $R^2 = 0.999996$, intercept/4 = 0.999984 ? OK

D6 v_F estimates (4): pass1 (7-pt fit ?72) = 1.8998, pass2 (11-pt mod-4 ?96) = 1.9447, pass3 (?=0.5 series ?80) = 1.955, pass4 (twist $L=48$) = 1.999

spread = 5.09% ? 6%, all in [1.85, 2.05] (analytic $v_F = 2t = 2$) ? OK

??

D1 twist-cone linear $R^2 > 0.995$ PASS ? $E_{\text{min}}(?) = 0.041645 \cdot v_F$ over 5 pts (no intercept), v_F twist = 1.9990, $n_{\text{zero}}(>0) = 0$??

D1b v_F twist vs 1/L-fit < 5% PASS ? twist 1.9990 vs pass-2 fit 1.9447 ? independent momentum-space measurement

D2 tower == 4 @1e-10 + index 2+2 PASS ? $\text{Tr}(P_A P_T) = 2.000000000$, $\text{Tr}(P_B P_T) = 2.000000000$? eigenvector-level AIII anchor

D3a 2D-holonomy identity PASS ? $\langle H \rangle \langle F \rangle \langle H \rangle = 2.5e-17$, $\max|E| = 4.4e-14$? Byers-Yang at the twist level

D3b ? cusp: tower 0, $E_{\text{min}} L$? 2 ? PASS ? $n_{\text{zero}}(?) = 0$, $E_{\text{min}} L = 6.2787$ (target 6.283, degeneracy 4)

D4 bit-exact rebuild PASS ? $\langle H \rangle_{\text{rebuild}} \langle H \rangle \langle F \rangle \langle H \rangle = 0.0e+00$? no hidden RNG/global state in the builder

D5 1/L ? collapse ? 4 ? PASS ? intercept 12.5662 (0.002% off 4?), $c = -82.06$, $R^2 = 0.999996$ over 11 pts

D6 cross-pass v_F ? 6% PASS ? 4 independent estimates, spread 5.09% ? twist + 3 fits agree

Test 19d [AUDIT pass 4]: 8 sub-checks, 0 failed ? PASS